

# Analogues of Theorems of Maizel And Pliss And Their Applications in The Shadowing Theory

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## The Concept of Shadowing

The theory of shadowing of approximate trajectories (pseudotrajectories) of dynamical systems is now a well-developed part of the global theory of dynamical systems. This theory is closely related to the classical theory of structural stability.

## Classical Shadowing

Let  $f$  be a diffeomorphism of a Riemannian manifold  $M$ .

**Definition.** A sequence  $\{x_k\}_{k \in \mathbb{Z}}$  of points of  $M$  is a *d-pseudotrajectory* if

$$\text{dist}(x_{k+1}, f(x_k)) < d, \quad k \in \mathbb{Z}.$$

**Definition.**  $f$  has *shadowing property* (pseudo-orbit tracing property,  $f \in \text{POTP}(= \text{StSh})$ ) if for a given  $\varepsilon$  there exists  $d > 0$  such that for any  $d$ -pseudotrajectory  $\{x_k\}$  there exists a point  $p \in M$ , such that

$$\text{dist}(x_k, f^k(p)) \leq \varepsilon, \quad k \in \mathbb{Z}.$$

We denote the set of all structurally stable diffeomorphisms of  $M$  by  $SS$ .

**Theorem (Sakai, 1994).**

$$SS = \text{Int}^1(\text{POTP}).$$

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**Definition.** A *decreasing d-pseudotrajectory* of the dynamical system, generated by  $f$  is a  $d$ -pseudotrajectory  $\{x_k\}_{k \in \mathbb{Z}}$ , such that

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**Theorem (Pilyugin, 2008).**

$$SS \subsetneq \text{TSLmSP},$$

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## Lipschitz Shadowing

**Definition.**  $f$  has *Lipschitz shadowing property* ( $f \in \text{LSP}$ ) if there exist constants  $L, d_0 > 0$  with the following property: for any  $d$ -pseudotrajectory  $\{x_k\}$  with  $d < d_0$  there exists a point  $p \in M$ , such that

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Lipschitz shadowing gives Lipschitz control on accuracy.

**Theorem (Pilyugin, Tikhomirov, 2010).**

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- Shadow this  $d$ -pseudotrajectory
- If  $d$  was small enough, you're done!

## Step 3. Shadowing $\Rightarrow$ solvability

One of the ways to obtain the structural stability is to use Mañé theorem.

For a point  $p \in M$  consider two linear subspaces of  $T_p M$  :

$$B^+(p) = \{v \in T_p M \mid |Df^k(p)v| \rightarrow 0, \quad k \rightarrow +\infty\}, \\ B^-(p) = \{v \in T_p M \mid |Df^k(p)v| \rightarrow 0, \quad k \rightarrow -\infty\}.$$

**Definition.** We say that  $f$  satisfy the *analytic transversality condition* at a point  $p$  if

$$B^+(p) + B^-(p) = T_p M.$$

**Theorem (Mañé, 1977).** Diffeomorphism  $f$  is structurally stable iff it satisfies analytic transversality condition for any  $p \in M$ .

## Step 2. Hyperbolicity $\Rightarrow$ stability

There is a relation between the solvability of the system of linear difference equations and some hyperbolicity properties of this system.

Let  $I$  be either  $\mathbb{Z}^+ = \{k \in \mathbb{Z} \mid k \geq 0\}$  or  $\mathbb{Z}^- = \{k \in \mathbb{Z} \mid k \leq 0\}$  or  $\mathbb{Z}$ .

Let  $\mathcal{A} = \{A_k\}_{k \in I}$  be a sequence of linear isomorphisms  $\mathbb{R}^d \rightarrow \mathbb{R}^d$  indexed by integers from  $I$ . Consider homogeneous and inhomogeneous equations associated with this sequence.

$$x_{k+1} = A_k x_k, \quad k \in I; \\ x_{k+1} = A_k x_k + f_{k+1}, \quad k \in I.$$

Fix  $\omega \geq 0$ . Denote the Banach space of sequences with bounded norm  $\|x\|_\omega = \sup_{k \in I} |x_k| (|k| + 1)^\omega$  by  $\mathcal{N}_\omega(I)$ .

Such spaces have been already studied in a similar context (Bichekguev, 2011).

**Definition.** A sequence  $\mathcal{A}$  has *Perron property*  $B_\omega(I)$  if  $\{f_k\}_{k \in \mathbb{Z}} \in \mathcal{N}_\omega(I) \Rightarrow$  solution of inhomogeneous equation  $\{x_k\}_{k \in \mathbb{Z}} \in \mathcal{N}_\omega(I)$ .

Such properties are often called *admissibility*.

## Hyperbolicity of a sequence

**Definition (Pilyugin, 2006).** We say that a sequence  $\mathcal{A}$  is hyperbolic on  $I$  if there exist constants  $K > 0, \lambda \in (0, 1)$  and projections  $P_k, Q_k, k \in I$  such that if  $S_k = P_k \mathbb{R}^d$  and  $U_k = Q_k \mathbb{R}^d$  then the following holds:

$$\mathbb{R}^d = S_k \oplus U_k; \\ A_k S_k = S_{k+1}, \quad A_k U_k = U_{k+1}; \\ |\Phi_{k,l} v| \leq K \lambda^{k-l} |v|, \quad v \in S_l, \quad k \geq l; \\ |\Phi_{k,l} v| \leq K \lambda^{l-k} |v|, \quad v \in U_l, \quad k \leq l; \\ \|P_k\|, \|Q_k\| \leq K.$$

Everywhere here we mean that all indices are from  $I$ .

## Theorems of Maizel and Pliss

**Theorem (a generalization of the discrete analog of Maizel Theorem).** Let  $I = \mathbb{Z}^+$  and the norms of all matrices  $A_k$  and  $A_k^{-1}$  be bounded by  $M > 0$ . A sequence  $\mathcal{A}$  has property  $B_\omega(I)$  iff it is hyperbolic on  $\mathbb{Z}^+$ .

**Theorem (a generalization of the discrete analog of Pliss Theorem).** Let  $I = \mathbb{Z}$  and the norms of all matrices  $A_k$  and  $A_k^{-1}$  be bounded by  $M > 0$ . A sequence  $\mathcal{A}$  has property  $B_\omega(I)$  iff it is hyperbolic on both  $\mathbb{Z}^+$  and  $\mathbb{Z}^-$  and the spaces  $B^+(\mathcal{A})$  and  $B^-(\mathcal{A})$  are transverse. Here

$$B^+(\mathcal{A}) = \{v \in \mathbb{R}^d \mid |\Phi_{k,0} v| \rightarrow 0, \quad k \rightarrow +\infty\}, \\ B^-(\mathcal{A}) = \{v \in \mathbb{R}^d \mid |\Phi_{k,0} v| \rightarrow 0, \quad k \rightarrow -\infty\}.$$

There are lots of papers concerning results of this type. They generalize theorems of Maizel and Pliss in different directions (Latushkin, Baskakov, Sasu). For infinite dimensional equations and/or different classes of spaces having certain homogeneity properties. Spaces  $\mathcal{N}_\omega(I)$  are neither homogeneous in the sense of Baskakov nor translation-Invariant in the sense of Sasu.

## New sort of shadowing

Having a method of obtaining structural stability and the generalisation of the Pliss' theorem, it is natural to introduce a new shadowing property to apply this generalisation.

Let  $\gamma \geq 0$ .

**Definition.** A sequence  $\{x_k\}_{k \in \mathbb{Z}}$  of points of  $M$  is a  *$\gamma$ -decreasing  $d$ -pseudotrajectory* if

$$\text{dist}(x_{k+1}, f(x_k)) < d(|k| + 1)^{-\gamma}, \quad k \in \mathbb{Z}.$$

**Definition.**  $f$  has *Lipschitz two-sided limit shadowing property with exponent  $\gamma$*  if there exist  $d_0, L > 0$  such that for any  $\gamma$ -decreasing  $d$ -pseudotrajectory  $\{x_k\}$  with  $d \leq d_0$  there exists a point  $p \in M$  such that

$$\text{dist}(x_k, f^k(p)) \leq Ld(|k| + 1)^{-\gamma}, \quad k \in \mathbb{Z}.$$

We write  $f \in \text{LTSLmSP}(\gamma)$  in this case.

**Theorem (Todorov, 2012).**

$$SS = \text{LTSLmSP}.$$